Program 40:Basic Mathematical Operations

```
%{
#include <stdio.h>
#include <stdlib.h>
%}

DIGIT   [0-9]+

%%

{DIGIT}[ ]*[\+\-\*/][ ]*{DIGIT} {
   int a, b;
   char op;

   sscanf(yytext, "%d %c %d", &a, &op, &b);

   switch(op)
   {
      case '+':
         printf("Result = %d\n", a + b);
         break;

      case '-':
         printf("Result = %d\n", a - b);
         break;

      case '*':
         printf("Result = %d\n", a * b);
         break;

      case '/':
```

```
            if(b == 0)
                printf("Error: Division by zero\n");
            else
                printf("Result = %.2f\n", (float)a / b);
            break;
    }
}

\n      { return 0; }

.       ;

%%
int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter expression (e.g., 10+5 or 20 / 4):\n");
    yylex();
    return 0;
}
```

Output:

```
C:\Users\SAIL\Downloads>gcc lex.yy.c -o math

C:\Users\SAIL\Downloads>math
Enter expression (e.g., 10+5 or 20 / 4):
20/4
Result = 5.00

C:\Users\SAIL\Downloads>
```